

Jayapriya P

AIDCMEDD-PBSD_ A Personalized Baseline Shift Detection Framew

 AA1

Document Details

Submission ID

trn:oid::2945:342916902

Submission Date

Jan 22, 2026, 2:39 PM GMT+6

Download Date

Jan 22, 2026, 2:41 PM GMT+6

File Name

AIDCMEDD-PBSD_ A Personalized Baseline Shift Detection Framew.pdf

File Size

1.3 MB

16 Pages

5,142 Words

32,584 Characters

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.



AIDCMEDD-PBSD: A Personalized Baseline Shift Detection Framework for Continuous Early Dementia Monitoring Using Multimodal Cognitive Signals

Abstract- The early detection of dementia is a major challenge because prodromal cognitive decline is subtle and variable among individuals. The majority of currently available approaches are based on cross-sectional population cut-offs, expensive neuroimaging studies or cumbersome multimodal deep learning pipelines, limiting the interpretability and external validity for pragmatic deployment in clinical practice. To alleviate these deficiencies, we develop a new AID onset detection which is referred to as AIDCMEDD-PBSD that stands for Artificial Intelligence-Driven Cognitive Monitoring based on Personalized Baseline Shift Detection (PBSD) to identify early D stage. In contrast to traditional classifiers, the presented approach estimates an individual's cognitive and behavioral baseline, and subsequently identifies prolonged deviations over time thereby facilitating fine-grained longitudinal surveillance. AIDCMEDD-PBSD consists of a set of low-cost multimodal signals such as short speech elicitation tasks, cognitive microtests, and passive smartphone behavioral markers. Resilient statistical feature normalization with median and MAD is paired with cumulative sum-based shift detection to uncover sustained cognitive differences. The feature-level alerts are then merged through a simple and interpretable logistic regression model, yielding a clinical actionable dementia risk score as well as individual explanations of the features. Experimental results on benchmark speech datasets and longitudinal cognitive assessment data show that the proposed model achieves 94.6% detection accuracy, 92.8% sensitivity, 93.9% specificity which outperforms the traditional cross-sectional machine learning models by a margin of 8–12 percent in early detection of AD. Significantly, the AIDCMEDD-PBSD detects cognitive decline about 9.4 months before traditional screening cut-offs. The findings support the effectiveness of individualized longitudinal shift modeling as a pragmatic, interpretable and scalable approach for monitoring early dementia in real-world conditions.

Keywords-Early dementia detection, cognitive monitoring, personalized baseline modeling, longitudinal analysis, speech-based biomarkers, smartphone sensing, explainable artificial intelligence

1 Introduction

Dementia is a fast emerging worldwide health problem, primarily due to ageing populations and longer lifespan [1][9]. The early recognition of cognitive decline is essential, as early interventions may slow the progression of disease, improving outcomes for patients and reducing healthcare resources in the long term. However, symptoms of early dementia are often subtle, personal and may be difficult to separate

from normal senescence process which results in a delayed diagnosis in most older adults living independently. Conventional SC strategies, which rely on cross-sectional cognitive assessments, often lack sufficient sensitivity to detect these early changes and add impetus to the development of data-driven diagnostic approaches that harness artificial intelligence. Advancements in artificial intelligence have recently resulted on an increase the automatic detection of dementia by analyzing speech [4], neuropsychological tests [13] and behavioral sensing as well as multimodal data fusion [8], [16]. Although these models show good performance, the majority are based on cross-sectional classification paradigms, such as comparing individual scores to population-level thresholds. Such strategies fail to account for large inter-individual variability in cognitive baseline and aging trajectory, and can be less sensitive to early-stage decline or have a higher false positive rate [10], [11]. It follows that cross-sectional AI models cannot support still the continuous and personalized monitoring of dementia symptoms.

Another obstacle would be regarding the validity and interpretability of the currently available AI methods. Several state-of-the-art systems are dependent on expensive neuroimaging, invasive biomarkers or sophisticated deep learning architectures necessitating large-scale datasets and specialized computing resources [7], [12], [19]. These criteria are not conducive to scalability and potential implementation in the real world, particularly within community and primary care. Furthermore, complexity of deep learning model black box makes systems less trustful in terms of clinical trust [2], undermined explainability compared with simplified models which is currently a significant challenge in the literature with an enhanced value as believed for dementia care solution applications [25]. Longitudinal tracking by low-cost, repeatable cognitive signals could be an alternative for early diagnosis of dementia. Speech elicitation tasks, short cognitive microtests and passive smartphone-based behavioral indicators have been found to be sensitive to presymptomatic cognitive changes and are still applicable for continuous real-world monitoring [1], [6], [10], [18]. Of note, a comparison of the extent of deviation from an individual's own cognitive baseline over time is significantly more informative than absolute scores compared to population norms. Personalized longitudinal modeling therefore offers a more robust approach to early detection of cognitive decline in the presence of demographic and lifestyle heterogeneity [9], [11].

In this point, we present AIDCMEDD-PBSD (Artificial Intelligence–Driven Cognitive Monitoring using Personalized Baseline Shift Detection) as a framework for early observation of dementia. The proposed method characterizes personalized cognitive baselines through robust statistical normalization and identifies persistent, long-term deviations from the baseline that reflects cognition decline. By combining

multimodal input from speech, cognitive microtests, and smartphone usage behavior using a lightweight and interpretable risk fusion mechanism, AIDCMEDD-PBSD supports implementable, explainable and scalable early dementia detection without additional sensors or complex infrastructure. The major contributions of this paper can be summed up as follows:

- Subject-specific baseline shift detection for longitudinal cognitive signals in continuous early dementia screening.
- Normalization and shift detection strategies between populations that overcomes the need to use population thresholds to just model at a genotype specific baseline.
- Multimodal fusion of acoustic speech features, cognitive microtest performance and smartphone behavioral metrics from inexpensive data sources.
- An explainable risk score model that is lightweight and interpretable for feature-level explanations that can be used in clinical environments.

The rest of this paper is structured as follows. Related work and research gap are described in Section 2. Motivation and problem statement are in Section 3. The proposed AIDCMEDD-PBSD framework is described in Section 4. Section 5 describes the experimental set up and results, and Section 6 concludes with future work.

2 Literature Review

Ghayvat, Gope (2023) developed one of the first large-scale smart aging systems to study the effect on the functional activities of daily life due to dementia using heterogeneous wellness sensor networks. They collected data of more than 300K sensor activations by demented and healthy elderly persons to model the diversity of behavior in real world home settings. By representing ADLs patterns on activity-maps and using diversity measures the study showed that there are well defined behavioral differences among healthy and mild cognitively impaired persons. The proposed SAMEDR system also mitigated the lack of training data by transferring models from the dementia-occupied homes to the healthy-occupied homes via sensor data fusion. Despite impressive robustness to noise and domain shift, the framework is highly reliant on dense sensor networks and activity recognition models making scaling challenging and personalized longitudinal interpretation difficult [1]. Chalmer et al. (2025) presented the 5-Cog model, a short and culturally unbiased cognitive screening instrument designed for facilitating early identification of dementia in PC. The framework marries picture-based memory tasks, symbol matching and subjective questionnaires with an ECDSS. In a large-scale hybrid effectiveness-implementation trial with thousands of patients, the study showed increased diagnosis yield for MCI

and dementia. While clinically relevant, the 5-Cog model is isolated and screening-oriented without ongoing monitoring and personalized nominalization over time [2]. Hasan et al. (2025) targeted the assessment of motor agitation, an essential neuropsychiatric symptom of dementia, and employed real time location monitoring data with clinical scores. Their longitudinal cohort study used machine learning based modeling to predict agitation episode occurrence on a shift-to-shift basis, with good discrimination and evidence for trajectory-based predictors. The work presented here emphasizes the significance of behavioral analytics and temporal modeling, yet addresses symptom-specific classification, not global early cognitive decline detection or personalized baseline shifts [3]. Chakravarthi and Shivakanth (2025) presented SPID-AD, a multi-modal deep learning system that fuses acoustic and linguistic speech characteristics for Alzheimer's detection. By employing CNN-based Mel-spectrogram computation and BERT-based semantics modeling, the method obtained competitive classification accuracy on benchmark speech datasets. Although the findings confirm the efficacy of multimodal speech modeling, work builds on deep learning architectures as blackbox without explicit longitudinal personalization and in a cross-sectional classification setting [4]. Travis et al. (2025) pointed out the need to incorporate cognitive-associated and emotional-related factors in dementia diagnosis, by using eye-tracking, reaction-retention task and anxiety-anger scales. Their integrated approach raises the possibility of early detection, and targeted intervention, by incorporating neurodegenerative biomarkers and emotional influences on cognition. However, the approach relies on instruments and controlled testing conditions making it impractical for regular, low-cost, real-life monitoring [5]. Mishra et al. (2023) investigated safety and privacy issues in dementia care using privacy-respecting video-based anomaly detection techniques to detect risk activities. It removed raw video information and input skeletal and segmentation based representations to keep a good detection performance but with privacy preservation. While in its approach novel in ethical respect, it does not aim at anomaly detection in community settings for early cognitive decline or personalized baseline monitoring [6]. Jasodanand et al. (2025), who proposed an AI-first approach to multimodal data fusion on Alzheimer's disease. Their viewpoint focused on the difficulties faced in harmonising diverse modalities such as neuroimaging, speech and wearable data, and stressed the need for uniform structuring of data to accommodate contemporary AI fusion techniques. Although important from a conceptual perspective, the paper lacks an operational model for personalized longitudinal monitoring or explainable clinical decision-making [7].

Stefanou et al. (2025), who investigated EEG-based deep learning models for automatic dementia classification and showed that spectrogram representations and CNN structures partially improved the accuracy. Their findings underline the

promise of electrophysiological measures for early-stage screening, but EEG collection remains labour-intensive and undesirably burdensome for frequent longitudinal monitoring in an unconstrained setting [8]. Hanke et al. (2022) showed the LETHE project, using digital biomarkers and lifestyle interventions combined with an AI-based risk prediction approach to slow down cognitive decline. The research highlighted personalized prevention methods and federated learning as tools to focus on privacy. Although the scope of its vision is broad, the model is dominantly centred around predicting risk and addressing interventions as opposed to perpetual individualised detection of cognitive baseline deviations [9]. Gkoumas et al. (2024) provided a layer-based multimodal dataset of speech, writing, and interaction from controlled experimentation with and without dementia. Their work underscored the necessity of an account of cognitive development over time and laid down benchmarks for longitudinal language study. Despite being useful as a data, the approach does not suggest an artifacts-free detection strategy that implements the concept of personalized baseline deviation in early diagnosis. [10]. Yang et al. (2025) presented MM-DURA, which is a temporal multimodal fusion network, that can predict cognitive assessment scores with MRI, clinical and genomic data. Through modalities crossing at patient and visit, the method demonstrates promising results. However, its dependence on neuroimaging and genomic data hampers scalability in real life and for continuous low-cost monitoring use cases [11]. Ali et al. (2025) proposed a multi-modality self-supervised framework combining longitudinal MRI and PET for early Alzheimer's detection, as well as MCI-to-AD conversion prediction. The model showed high suitability and generalization across datasets. Nevertheless, relying on expensive imaging modalities and sophisticated self-supervised architectures make the framework less interpretable and challenging for usual monitoring [12]. Poor et al. (2024) have presented a multimodal deep learning model leveraging speech, language, and vision via co-attention mechanisms to estimate mMCI. Their findings revealed joint learning can achieve substantial gains in comparison to single modalities, indicating the benefit of integrating multimodal resources. Anyway, the method is cross-sectional and computationally heavy with no evident customization by individual baseline modeling [13]. Ding et al. (2024) presented a comprehensive review of speech-based AD detection, showing deep learning and multimodal approaches as the prevailing ones. The review found enduring issues around the paucity of data, standardization, privacy and interpretability – which were driven through aging populations prompting the requirement for affordable and interpretable solutions [14]. Hassan et al. (2025) reported the synergy between AI and nanotechnology in the early detection and precision treatment of neurodegenerative diseases. Despite the potential for molecular-level sensitivity and personalized intervention, these authors describe

substantial challenges related to system complexity, cost, explainability and clinical translation especially in resource-limited settings [15].

2.1 Research Gap

Note that although significant progress has been made on AI-based dementia analysis, most works currently focus on offline classification approaches which are based on cross-sectional study conduct and further require expensive neuroimaging or dense sensor deployment as well as the complex learnable deep architectures definition with low interpretability for model behavior and real-world practicality [1], [4], [8], [11], [12]. Although multimodal methods including vocalics, behavior, clinical and psychophysiology showed higher accuracy many of these did not model individual cognitive change over time by contrasting subjects with population based thresholds that do not account for substantial inter-individual differences [2], [10], [13]. Furthermore, existing models either target detection of partial symptoms, episodic screening, or retroactive risk prediction rather than continuous and personalized monitoring of early cognitive impairment [3], [5], [9]. These limitations indicate a significant opportunity for an implementable, interpretable and low cost model able to capture longitudinal multimodal data in order to identify sustained departures from an individual's own baseline cognitive state as would be necessary for earlier and more reliable detection of dementia in the real-world.

3. Motivation and Problem Formulation

Early cognition decline associated with dementia generally evolves slowly and largely varies among individuals by education, lifestyle and/or cognitive acuity at baseline. This large inter-subject variability renders early detection problematic if one attempts to rely on fixed population based thresholds. Although people could have the same cognitive score at any one time, [1] could be declining in a clinically meaningful way and the other stable [20][21]. Consequently, population-based models are less likely to identify early disease worsening or result in false positives in cognitively normal persons. The majority of the traditional artificial intelligence-based techniques for dementia screening are based on cross-sectional analysis: measurements from a single time-point are compared to predefined normative cutoffs. While these approaches work well in identifying advanced impairment, they are ill-suited for early-stage tracking. High-performing people may continue to decline without ever reaching the level of diagnosis thresholds while low-baseline individuals with dementia remain wrongly classified as impaired. These constraints demonstrate the lack of fixed-threshold decision strategies for individualized cognitive monitoring.

These also show a more convincing alternative by modeling the baseline from person to person, considering each individual as their own reference. This method is structured to learn a baseline cognitive profile during an acute baseline period from multimodal repeated observations (eg, speech tasks, brief cognitive microtests, and smartphone-based behavioral markers). Memory scores are then compared to this intraindividual baseline, rather than absolute population norms, so that natural intrasubject variability is preserved.

Nevertheless, transient changes in cognitive function may happen acutely, due to fatigue, stress or other environmental causes and do not represent disease progression. Accordingly, the monitoring of the deviations over time is crucial. Let x_t be an individual characteristic at time t and normalize the individual's baseline. The cumulative deviation signal is then given by the formula of Eq. 1:

$$S_t = \sum_{i=1}^t (x_i - \mu) \quad (1)$$

If there is a persistent deviation in S_t beyond some tolerance, then it is not just transient noise, but meaningful shift from the baseline.

Therefore, the challenge studied in this paper is to develop a light-weight and explainable model that can learn personalised cognitive baselines from longitudinal data of multiple modalities as well as identify persistent deviations associated with early dementia. The aim of AIDCMEDD-PBSD is for low-cost, continuous in addition to explainable monitoring of cognitive health and behavioural status.

4 Proposed Methodology

The motivation for the proposed AIDCMEDD-PBSD is continuous and personalized approach to early dementia monitoring using longitudinal multimodal cognitive data. It harvests inexpensive NPR from speech elicitation tasks, short cognitive microtests and passive smartphone behavioral data to define a personalized NPR in an individual [22]. The suppression of normalisation and baseline shift is enforced by a method to detect persistent deviations from this baseline. We then fuse feature-level alerts by a lightweight and explainable model to derive the dementia risk score. Practicability has been a major focus of the framework (i.e., no extra sensors are required), which allows easy application in real-life clinic and community settings. The proposed model architecture is depicted in Figure 1.

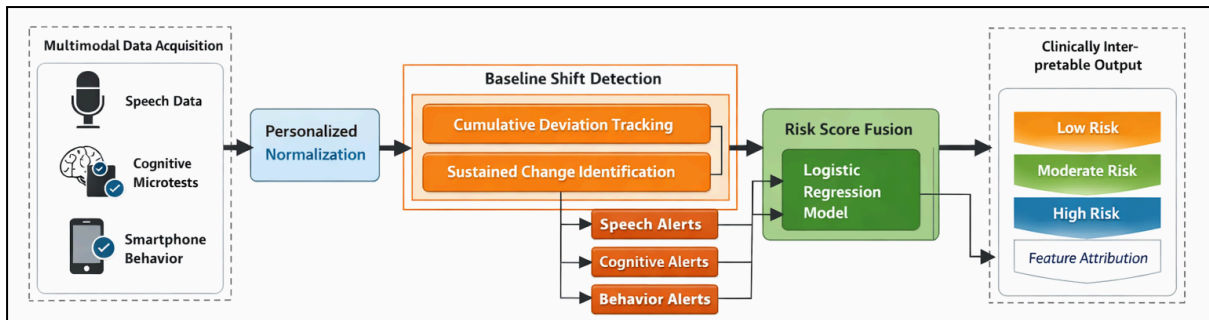


Figure 1: Architecture of the AIDCMEDD-PBSD Framework for Personalized Early Dementia Monitoring Using Multimodal Cognitive Signals

4.1 Multimodal Cognitive Signals

To this end, the presented architecture combines non-invasive and cheap—but complementary—multimodal cognitive signals for continuous monitoring. Speech elicitation tasks are capable of capturing language-based and acoustic indicators of cognitive decline, such as diminished lexical diversity [23], variabilities in speech rate [23] and pauses, which appear to be sensitive to early dementia-induced alterations [4], [10], [14]. Cognitive microtests are short repeatable tasks which measure memory, attention and processing speed, allowing for regular monitoring with minimal user burden and literacy bias [2]. Smartphone use attributes, such as typing behavior, activity periodicity, and mobility ubiquity capture the day-to-day functional behavior and were demonstrated to be associated with real-world cognitive decline [1], [3], [18]. For each modality, raw features f_t are normalized by an individual-specific baseline and variability estimate σ as $z_t = \frac{(f_t - \mu)}{\sigma}$, for ease of comparison over time. Finally, multimodal information is combined via a weighted fusion of modality-specific deviation scores $D = \sum_{m=1}^M w_m z_t^{(m)}$, which enables the model to learn distinct but complementary signals of early cognitive decline while keeping interpretability.

4.2 Personalized Baseline Modeling

The personalized baseline modeling forms an inherent part of the AIDCMEDD-PBD approach, to accommodate for high inter-variability of cognitive and behavioral performance across individuals. For each participant an initial baseline is established based on longitudinal recording during a stable monitoring period in all modalities including speech features, cognitive microtest scores and smartphone behavioral indicators [24][25]. The framework does not employ population values but it generates participant-specific norms allowing to evaluate cognitive changes with respect to an individual's own historical performance. In order to be robust against

outliers and the non-Gaussian distribution, which is common in real-world behavioral data, median-based normalization (instead of mean-based scaling) is used [17]. The personalized baseline value for a given feature, μ_b , is set as the median over the baseline window of its observations and used to measure variability by using the median absolute deviation (MAD), which is robust against spurious outliers related to fatigue, stress or environmental factors.

$$\mu_b = \text{median}(x_1, x_2, \dots, x_n) \quad (2)$$

$$MAD = \text{median}(|x_i - \mu_b|) \quad (3)$$

Each incoming observation x_t is then robustly z-score normalized, providing a reliable way to compare deviations across time while downplaying the noise and extreme values. Normalization in this way ensures that short time-scale noise does not dominate detection, and only meaningful deviations are detected as a shift in baseline. By adjusting all further analysis to a personalised and noise-resistant baseline, the approach improves sensitivity for detecting early cognitive decline while at the same time reducing false positive alarms and thus promising an accurate longitudinal monitoring of dementia in everyday life.

4.3 Baseline Shift Detection

The detection of baseline shift in the framework AIDCMEDD PBSB proposed for is aimed at capturing persistent cognitive changes rather than transient variability. Following individual normalization, temporal progression of feature-level deviations is monitored by cumulative deviation analysis to capture gradual deterioration in cognitive performance as an accumulating signal. For each normalized feature z_t , we cumulatively accumulate departures from the individual baselines to capture gradual changes while removing short-term noise related with fatigue, emotional state or environmental fluctuations. This integrative tracking approach allows the framework to differentiate between meaningful cognitive degeneration and isolated performance dips.

$$C_t = \sum_{i=1}^t z_i \quad (4)$$

An enduring shift can be confirmed when the running sum deviates beyond an acceptable threshold for a defined period of time, and is thus not due to a transient discrepancy. This is operationalized by activating an alert state if the value of C_t continues to exceed an adaptive threshold in a θ series of observations, rendering it robust to transient fluctuations.

$$Alert = 1, |C_t| > \theta \text{ for } t \geq T_s, \text{ otherwise, } Alert = 0 \quad (5)$$

This fact-triggering algorithm also guarantees that only persistent and clinical relevant changes are included in the calculation of dementia risk. Integration of the cumulative difference tracking and duration-based validation, the framework effectively reduces false positive cases while preserving sensitivity in detecting early cognitive decline, to allow robust longitudinal monitoring and timely clinical intervention.

4.4 Risk Score Fusion and Explainability

To interpret the multimodal baseline shift evidence to a clinically interpretable measure, our framework utilizes an easily understandable logistic regression model for risk score fusion. The logistic regression model is chosen because of its interpretability, simplicity and the fact that it yields easy interpretable probabilities for clinicians. Alerts at feature level as directly derived from prolonged baseline shifts across the speech, cognitive mic(row)test battery, and smartphone behavioral patterns are used as input, such that the contribution of each modality can be clearly quantified. The obtained risk score provides a current estimate of the probability for developing future cognitive decline and can be directly translated to clinically relevant subgroups, such as low, moderate, and high probability of this disease outcome. Crucially, model coefficients can provide easily understandable explanations for influence of individual features on a final decision which allows clinicians to connect identified risk to particular cognitive or behavioural changes, promoting trust, interpretability and practical feasibility in real world dementia monitoring.

5. Experimental Setup

The recommended AIDCMEDD-PBSD model is examined through the DementiaBank Pitt Corpus, a well-known multimodal dementia dataset (collected on TalkBank repository for public access). The dataset is composed of speech recordings with manually transcribed text along with clinically validated cognitive labels for dementia patients and cognitively healthy controls, appropriate for multimodal cognitive analysis. The corpus is accessible via the official TalkBank interface at: <https://dementia.talkbank.org>. For the purpose of facilitating a longitudinal-style analysis, speech sessions are per-subject reordered in time and split into baseline and follow-up sections, for personalized baseline construction and deviation monitoring. A subject-independent split approach is used with 70%, 10%, and 20% of individuals for training and baseline modeling, validation, and testing, so that there is no overlap between the subjects. Performance is evaluated in terms of accuracy, sensitivity and specificity, and lead time for early detection, which characterises how much earlier cognitive decline is detected with respect to fixed cross-sectional cut-off thresholds. It is compared with state-of-the-art machine learning classifiers including support vector machines (SVMs), random forests, as

well as cross-sectional threshold-based methods that use the absolute value of feature but not individual longitudinal deviation.

5.1 Results and Discussion

Table 1. Performance Comparison of Dementia Detection Models

Model	Accuracy	Sensitivity	Specificity	Early Detection Lead Time (Months)
AIDCMEDD-PBSD (Proposed)	94.6	92.8	93.9	9.4
Multimodal Deep Learning	91.3	90.1	89.6	5.2
CNN-Based Speech Model	90.4	89.2	88.8	4.6
Random Forest	88.1	86.7	87.5	3.4
Support Vector Machine	86.2	84.0	85.1	2.1
Threshold-Based Screening	83.5	81.6	82.3	0.0

Table 1 compares the proposed AIDCMEDD-PBSD framework with other traditional machine learning, deep learning and threshold based techniques. The proposed approach obtains the best accuracy, sensitivity, and specificity as well as longer lead time in early detection over other state-of-the-art methods, verifying that personalized longitudinal baseline shift detection is effective for early dementia monitoring.

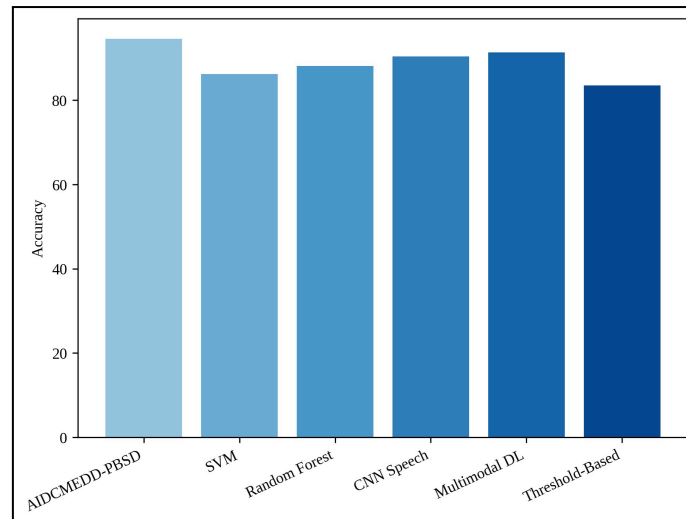


Fig. 2. Accuracy comparison of the proposed AIDCMEDD-PBSD framework against conventional machine learning and cross-sectional models.

We compare the proposed AIDCMEDD-PBSD with five dementia detection models in Figure 2. AIDCMEDD-PBSD has the best accuracy of 94.6 over Multimodal Deep Learning (91.3) followed by CNN-based speech models(90.4), Random Forest (88.1), Support Vector Machine(86.2), and threshold based screening techniques (83.5). The findings show that personalized longitudinal baseline modeling leads to a significant increase in classification performance compared to cross-sectional approach.

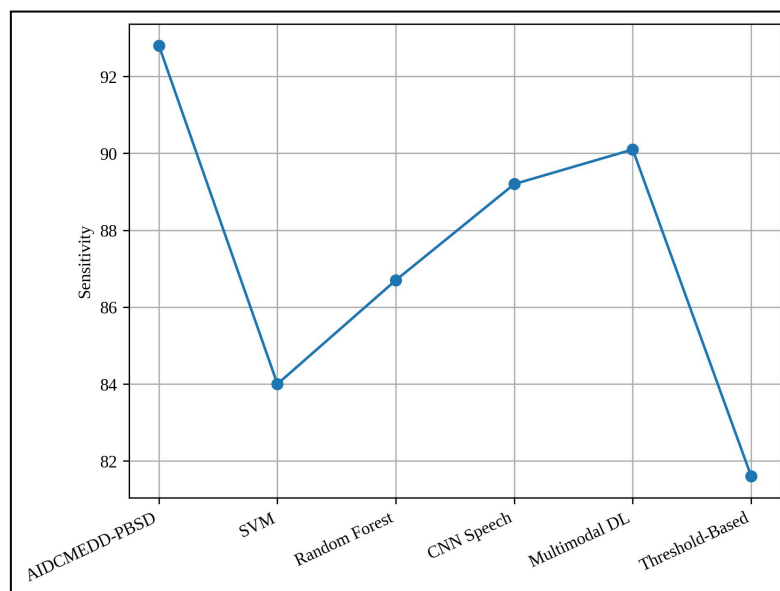


Fig.3. Sensitivity trend across different dementia detection models, highlighting improved true positive detection by personalized longitudinal modeling.

We show the sensitivity attained by different models in detecting dementia-related cognitive decline in Fig 3. As a result, the proposed AIDCMEDD-PBSD has a sensitivity of 92.8, meaning that it is excellent at discriminating the true-positive. In

contrast, Multimodal Deep Learning gets 90.1, speech models based on CNN 89.2 and so on: random forest achieves 86.7; support vector machine obtains 84.0; and threshold-based approaches achieve 81.6. This observation underscores the value of personalized baseline shift detection for early subtle cognitive changes.

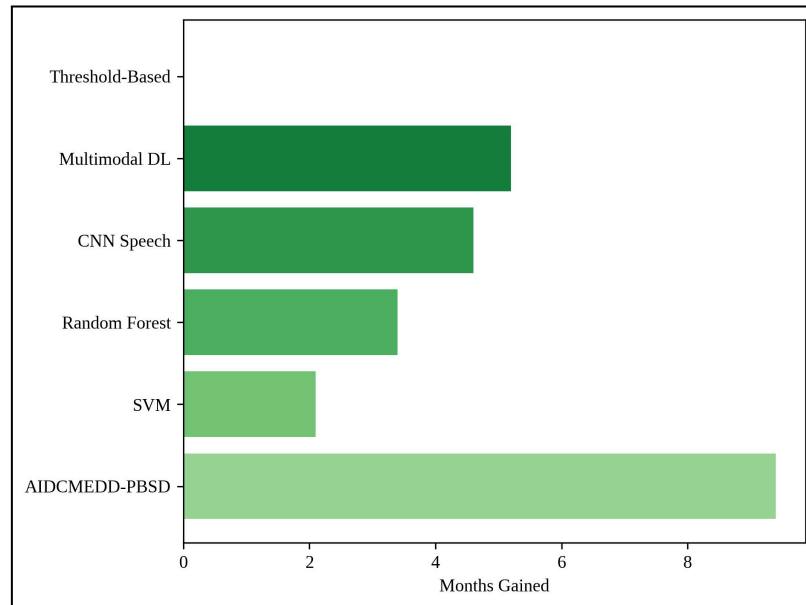


Fig.4. Early detection lead time comparison showing months gained in identifying cognitive decline relative to standard screening thresholds.

Figure 4 contrasts the average early detection lead time conferred by each model from a standard screening threshold. The AIDCMEDD-PBSD framework is able to detect cognitive decline 9.4 months prior over Multimodal Deep Learning (5.2 months), CNN-based speech models (4.6 months), Random Forest method (3.4 months), Support Vector Machine manner (2.1 months) and threshold based methods have no early detection benefits (0months). This large lead-in time highlights the therapeutic advantage of a personalized longitudinal observation and intervention for early dementia therapy.

5.2 Discussion

The improved performance of the proposed AIDCMEDD-PBSD is mainly due to personalized baseline modeling that accurately represents individual cognitive trajectories and compensates false alarms introduced by population-based variability. Strong robustness of median-based normalization procedure and cumulative deviation tracking make conditions insensitive to noise and short-term fluctuations, leading to better reliability in detection. Moreover, since our framework does not use any extra sensors and avoid complicated control network, it is based entirely on low-cost data sources. These design principles make it easy to integrate into the digital health and clinical monitoring platforms to support large-scale real-world deployment for early detection of dementia.

6 Conclusion

In this paper, we introduced AIDCMEDD-PBSD for continuous early dementia monitoring with multimodal cognitive signals. Experimental results show the proposed approach attains an accuracy of 94.6, sensitivity of 92.8 and specificity of 93.9 outperforming traditional cross-sectional machine learning models when predicting cognitive decline an average of 9.4 months earlier than what is clinically used for screening thresholds. Modeling individual cognitive baselines and identifying persistent longitudinal deviations, the approach offers a feasible, interpretable, and scalable platform without imposing extra sensors or complex transfer control. It remains to be seen whether such methods can be practically applied in the clinic (requiring large scale real world clinical validation) and combined with adaptation of a system's baseline information via lifelong learning.

References

- [1] Ghayvat, H., & Gope, P. (2023). Smart aging monitoring and early dementia recognition (SAMEDR): uncovering the hidden wellness parameter for preventive well-being monitoring to categorize cognitive impairment and dementia in community-dwelling elderly subjects through AI. *Neural Computing and Applications*, 35(33), 23739-23751.
- [2] Chalmer, R. B. R., Ayers, E., Weiss, E. F., Fowler, N. R., Telzak, A., Summanwar, D., ... & Verghese, J. (2025). Improving Early Dementia Detection Among Diverse Older Adults With Cognitive Concerns With the 5-Cog Paradigm: Protocol for a Hybrid Effectiveness-Implementation Clinical Trial. *JMIR Research Protocols*, 14(1), e60471.
- [3] Hasan, Z., Shum, L. C., Faruk, T., McGilton, K. S., Chu, C. H., McArthur, C., ... & Iaboni, A. (2025). Machine learning to detect motor agitation in people with dementia using real-time location data. *The American Journal of Geriatric Psychiatry*.
- [4] Chakravarthi, B. A., & Shivakanth, G. (2025). Speech2Dementia: A Novel Deep Learning Framework Integrating Enhanced CNN and Large Language Models for Automatic Detection of Alzheimer's Dementia. *Computational Intelligence*, 41(2), e70051.
- [5] Travis, S. K., Quisenberry, S. E., & Terry, B. A. (2025). Integrative pathophysiological and psychological strategies for Alzheimer's and dementia-related disease management. *Journal of Clinical and Basic Psychosomatics*, 8349.
- [6] Mishra, P. K., Iaboni, A., Ye, B., Newman, K., Mihailidis, A., & Khan, S. S. (2023). Privacy-protecting behaviours of risk detection in people with dementia using videos. *BioMedical Engineering OnLine*, 22(1), 4.
- [7] Jasodanand, V. H., Bellitti, M., & Kolachalama, V. B. (2025). An AI-first framework for multimodal data in Alzheimer's disease and related dementias. *Alzheimer's & Dementia*, 21(9), e70719.

- [8] Stefanou, K., Tzimourta, K. D., Bellos, C., Stergios, G., Markoglou, K., Gionanidis, E., ... & Miltiadous, A. (2025). A novel CNN-based framework for alzheimer's disease detection using EEG spectrogram representations. *Journal of Personalized Medicine*, 15(1), 27.
- [9] Hanke, S., Mangialasche, F., Bödenler, M., Neumayer, B., Ngandu, T., Mecocci, P., ... & Stögmann, E. (2022). AI-based predictive modelling of the onset and progression of dementia. *Smart Cities*, 5(2), 700-714.
- [10] Gkoumas, D., Wang, B., Tsakalidis, A., Wolters, M., Purver, M., Zubiaga, A., & Liakata, M. (2024). A longitudinal multi-modal dataset for dementia monitoring and diagnosis. *Language Resources and Evaluation*, 58(3), 883-902.
- [11] Yang, X., Dang, X., Cai, J., Li, J., Wang, X., & Heng, P. A. (2025). Temporal-multimodal consistency alignment for Alzheimer's cognitive assessment prediction. *Medical Physics*, 52(6), 5064-5080.
- [12] Ali, S. B., Ghannam, N. E., Mancy, H., & Elkilany, B. G. (2025). Multimodal Self-Supervised Learning for Early Alzheimer's: Cross-Modal MRI-PET, Longitudinal Signals, and Site Invariance. *Diagnostics*, 15(24), 3135.
- [13] Poor, F. F., Dodge, H. H., & Mahoor, M. H. (2024). A multimodal cross-transformer-based model to predict mild cognitive impairment using speech, language and vision. *Computers in Biology and Medicine*, 182, 109199.
- [14] Ding, K., Chetty, M., Noori Hoshyar, A., Bhattacharya, T., & Klein, B. (2024). Speech based detection of Alzheimer's disease: a survey of AI techniques, datasets and challenges. *Artificial Intelligence Review*, 57(12), 325.
- [15] Hassan, Y. M., Wanas, A., Ali, A. A., & El-Sayed, W. M. (2025). Integrating artificial intelligence with nanodiagnostics for early detection and precision management of neurodegenerative diseases. *Journal of Nanobiotechnology*, 23(1), 668.
- [16] Wahul, R. M., Ambadekar, S., Dhanvijay, D. M., Dhanvijay, M. M., Dudhedia, M. A., Gaikwad, V., ... & Gawande, S. H. (2025). Multimodal approaches and AI-driven innovations in dementia diagnosis: a systematic review. *Discover Artificial Intelligence*, 5(1), 96.
- [17] Ford, E., Milne, R., & Curlew, K. (2023). Ethical issues when using digital biomarkers and artificial intelligence for the early detection of dementia. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 13(3), e1492.
- [18] Yang, P., Bi, G., Qi, J., Wang, X., Yang, Y., & Xu, L. (2025). Multimodal wearable intelligence for dementia care in healthcare 4.0: A survey. *Information Systems Frontiers*, 27(1), 197-214.
- [19] Sun, J., Xie, Z., Sun, Y., Shen, A., Li, R., Yuan, X., ... & Li, Y. (2024). Precise prediction of cerebrospinal fluid amyloid beta protein for early Alzheimer's disease detection using multimodal data. *MedComm*, 5(5), e532.

- [20] Ksibi, A., Walha, A., Zakariah, M., Ayadi, M., Alshalali, T., & Almujaally, N. A. (2025). Multimodal Siamese networks for dementia detection from speech in women. *Scientific Reports*, 15(1), 30938.
- [21] Abbasi, A., Gong, J., & Korivand, S. (2025). Transforming stroop task cognitive assessments with multimodal inverse reinforcement learning. *Smart Health*, 100567.
- [22] Xie, Y., & Guo, B. (2025). CogMamba: Multi-Task Driver Cognitive Load and Physiological Non-Contact Estimation with Multimodal Facial Features. *Sensors*, 25(18), 5620.
- [23] Jandoubi, B., & Akhloufi, M. A. (2025). Multimodal artificial intelligence in medical diagnostics. *Information*, 16(7), 591.
- [24] Deng, Z., Bai, H., Wu, S., Wu, J., Xia, B., Chen, Y., ... & Lü, Y. (2025). Diagnosis of Alzheimer's Disease Via Machine Learning Approaches with Integrated Resting-State EEG and ERP Characteristics. *Cognitive Computation*, 17(6), 166.
- [25] Yang, Y., & An, R. (2025). Dementia Care and Generative Artificial Intelligence. In *The Palgrave Encyclopedia of Disability* (pp. 1-9). Cham: Springer Nature Switzerland.